

Gravitational Wave Event Dashboard

Gravitational wave astronomy has grown from 3 detections in its first 4-month run to over 100 in a single year, with BBH mergers consistently comprising over 70% of all events across every observing run

265

Total Events

61.17

Avg Mass

175

BBH Events

0.66

BBH %

11.52

Avg SNR

event_type

☐ BBH

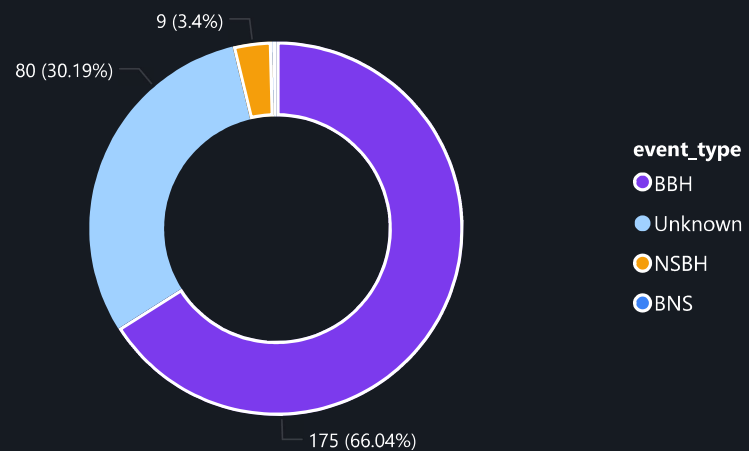
☐ BNS

☐ NSBH

☐ Unknown

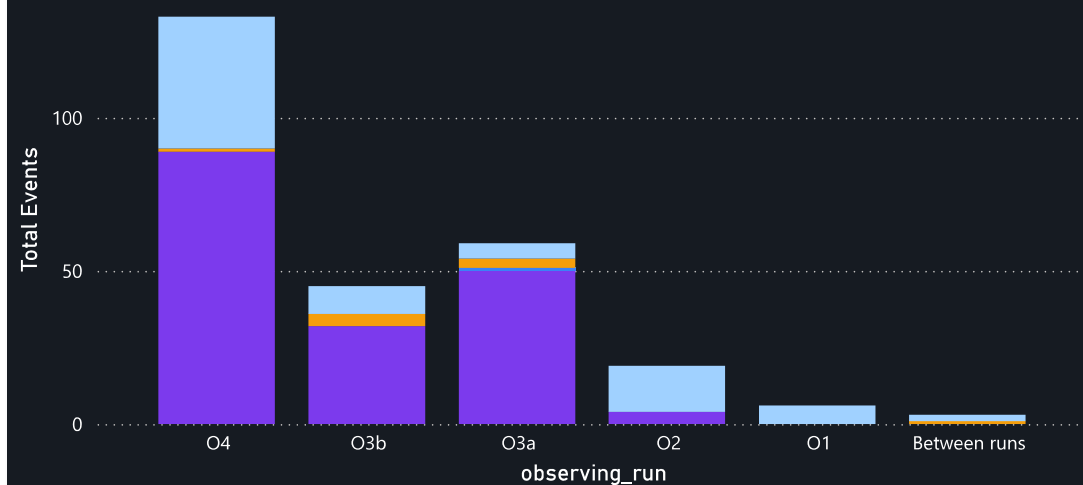
Total Events

by event_type



Total Events by observing_run and event_type

event_type ● BBH ● BNS ● NSBH ● Unknown

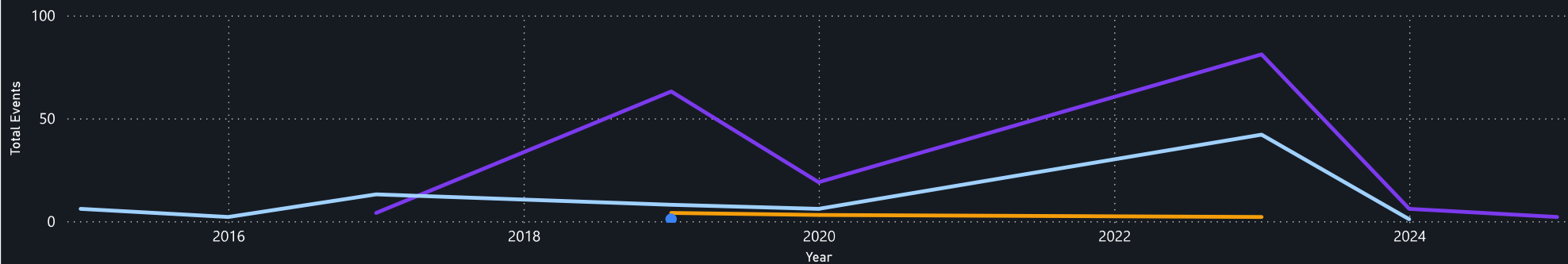


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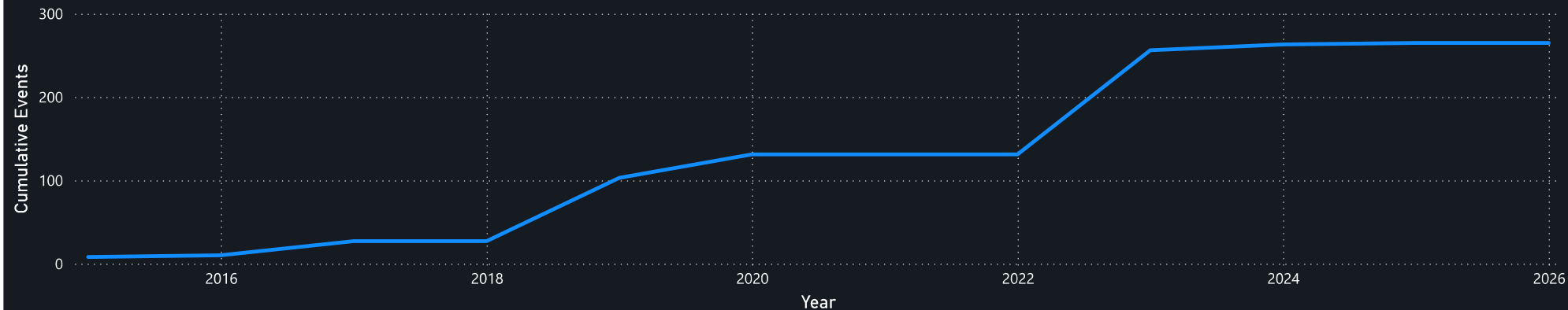
Detection capability has not grown linearly, it has grown in step-changes tied to hardware upgrades. Each observing run represents a distinct generation of detector sensitivity, and the gaps between runs are as analytically important as the runs themselves.

Total Events by Year and event_type

event_type ● BBH ● BNS ● NSBH ● Unknown



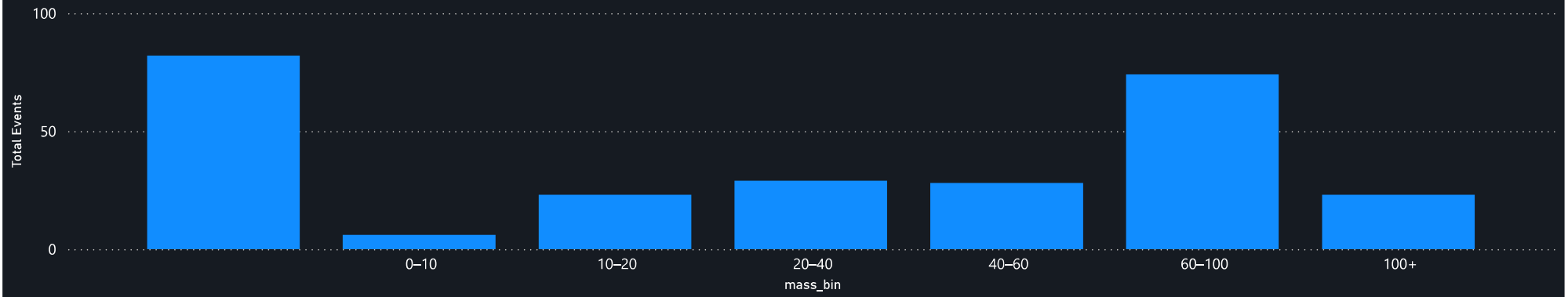
Cumulative Events by Year



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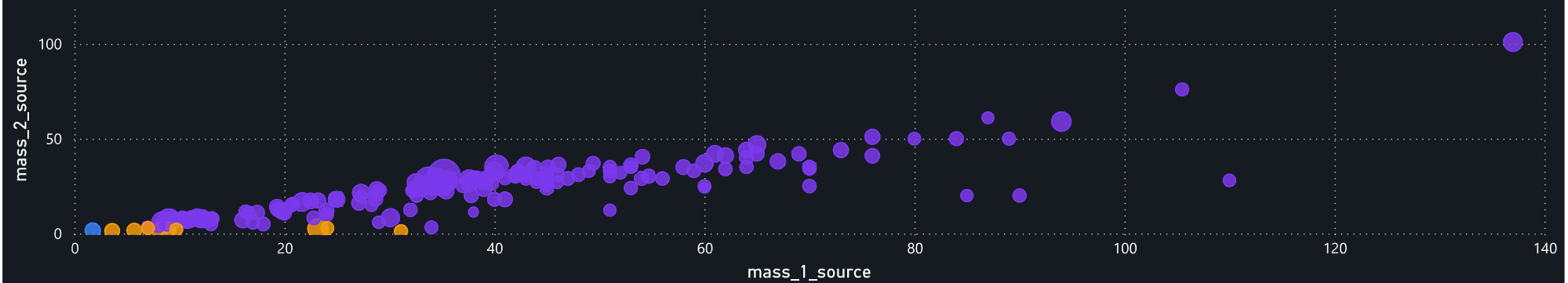
The mass distribution is bimodal — there is a cluster of low-mass events (neutron stars, 1–3 solar masses) and a broad distribution of higher-mass events (black holes, 5–150 solar masses). The gap between them around 3–5 solar masses — the so-called 'mass gap' — has only a handful of candidates, making those events scientifically extraordinary.

Total Events by mass_bin



Sum of snr by event_type, mass_1_source and mass_2_source

event_type ● BBH ● BNS ● NSBH ● Unknown



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LIGO's detection horizon - the maximum distance at which it can reliably detect a merger — has expanded from roughly 400 Mpc in O1 to over 4,000 Mpc in O4. This 10x increase in reach translates to a roughly 1,000x increase in the volume of space surveyed, directly explaining the exponential growth in event counts.

9K

Farthest Event

11.52

Avg SNR

6

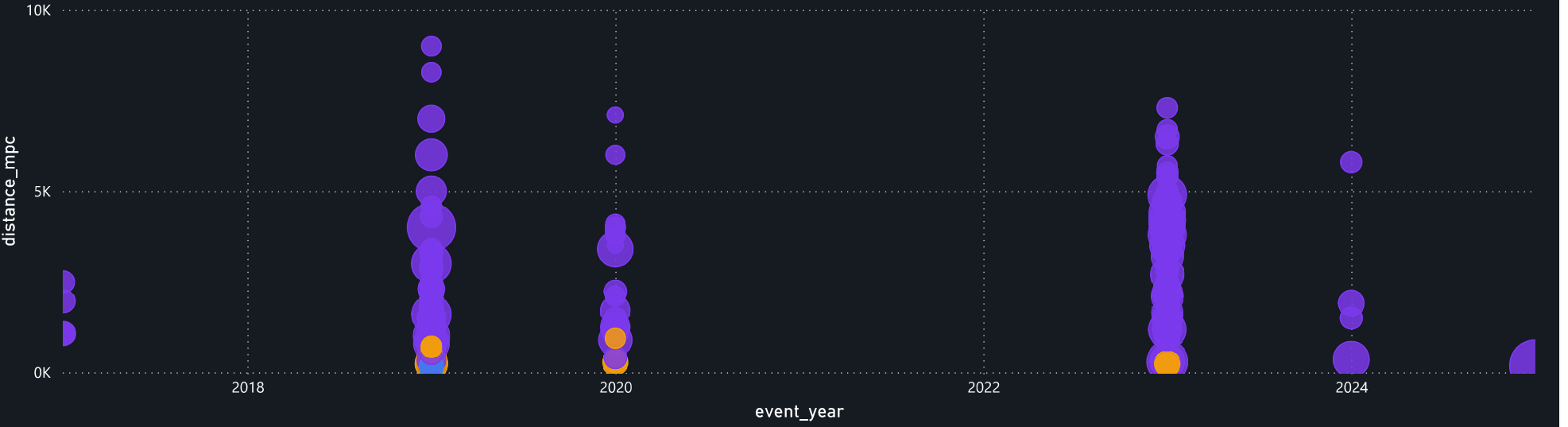
Total Runs

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Detection Growth

Sum of snr and First commonName by event_type, event_year and distance_mpc

event_type ● BBH ● BNS ● NSBH ● Unknown

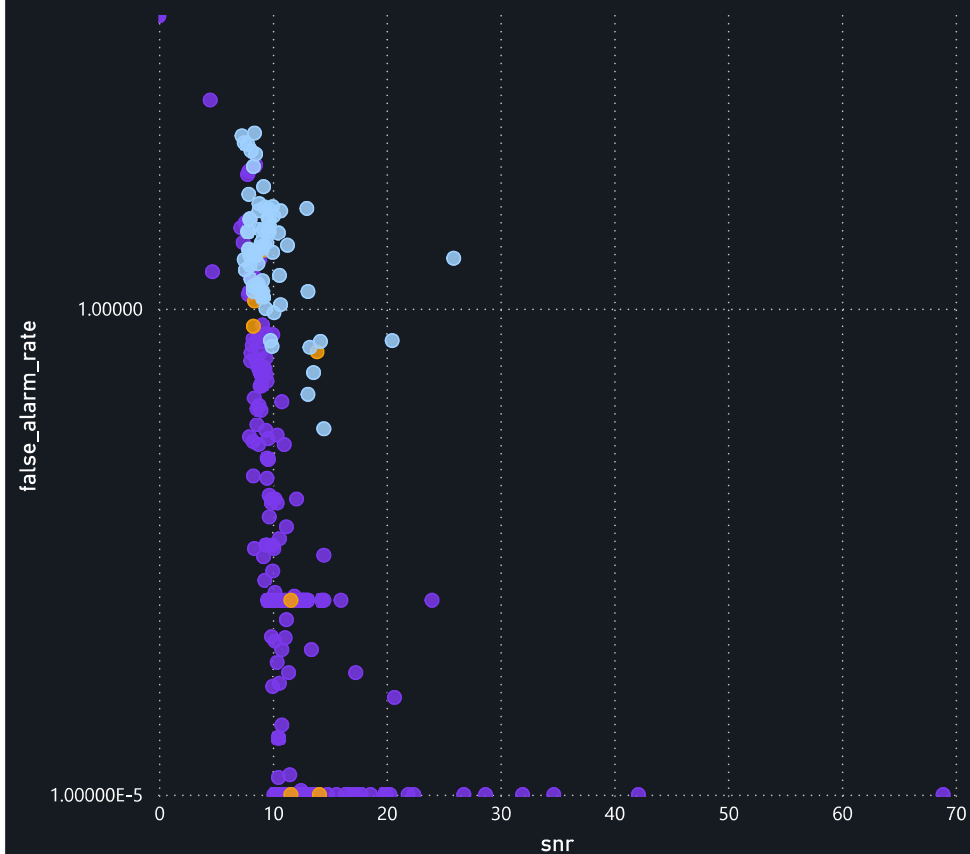


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Higher signal-to-noise detections consistently correspond to lower false alarm rates, reinforcing the reliability of confirmed gravitational wave events.

event_type, snr and false_alarm_rate

event_type ● BBH ● BNS ● NSBH ● Unknown



Sum of snr by commonName

